SOCIAL MEDIA ENGAGEMENT ANALYTICS USING PYTHON

Project Report

1. Project Objective

- Analyse social media engagement using Python.
- Explore content performance, audience behaviour, user interactions and sentiment patterns.

2. Dataset Overview

- Dataset size: **5,000 records and 19 original columns**.
- Includes user details, post type and category, likes, comments, shares, watch time, impressions, follower count, posting date, device type, sentiment, hashtags and engagement rate.

3. Data Cleaning and Preparation

- Identified and handled 900 missing values.
- Used median imputation for numerical columns and mode imputation for categorical columns.
- Checked for duplicate records; no duplicate rows were found.
- Standardised categorical text and converted posting dates to datetime format.
- Cleaned hashtag text and created a hashtag-count feature.
- Created an engagement score using likes, comments and shares.
- Final cleaned dataset: **5,000 records with no missing values or duplicate rows**.

4. Statistical Analysis

- Calculated mean, median and mode.
- Calculated standard deviation, variance and percentiles.
- Analysed likes, comments, shares, watch time, engagement rate and follower count.
- Compared engagement across post types, categories, countries and sentiment groups.

5. Visualization

- Used Matplotlib and Seaborn for distribution, comparison, relationship and correlation charts.
- Used Plotly to create interactive line, bar, scatter and bubble charts.
- Visualized daily engagement, post categories, sentiment, device type and audience-related patterns.

6. Key Findings

- Video posts recorded the highest average engagement rate among post types (approximately **1.12**).
- Negative-sentiment posts had an average engagement rate of approximately **1.04**, followed by neutral (0.99) and positive (0.92) posts.
- Age groups, verified-account status, posting hours and device types were also examined to understand differences in engagement and user behaviour.

7. Conclusion

- The project demonstrates a complete data-analysis workflow using Python, from cleaning and feature creation to statistical analysis and visualization.
- The findings provide useful insights into social media interactions and can support more informed content-planning decisions.

Tools and Libraries: Python, Pandas, NumPy, Matplotlib, Seaborn and Plotly.